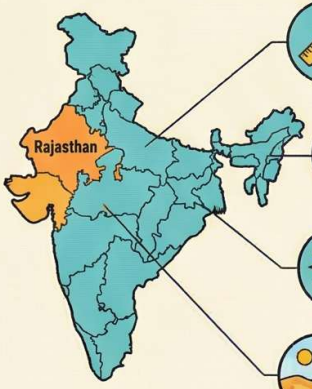
# — BHADLA SOLAR PARK — RENEWABLE ENERGY MANAGEMENT

RAJASTHAN, INDIA | WORLD'S LARGEST OPERATIONAL SOLAR PARK

**Case Study — Chinmay J Pagare (22108A0022)**

## PROJECT AT A GLANCE




Land Area: **14,000 acres**

Annual Generation: **~7.2 Billion kWh/year**

Solar Irradiance: **~5.72 kWh/m²/day**

Location: **Thar Desert**

## ENERGY MANAGEMENT APPROACH

Total Capacity: **2,245 MW**

Households Powered: **~6 Million**

**Smart Solar Tracking**

Rotating panels rotating solar panels.

**Real-time SCADA Monitoring**

Real-time SCADA Monitoring wnen managoness to exploer trackon panmels.

**Smart Grid Integration**

Smart grid integration to oyions and omomation.

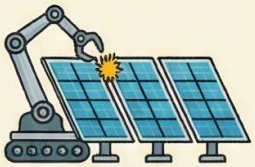
**Load Balancing**

Load balancing enences with scale panels in energy ambols.

## CHALLENGES & RESPONSES

| | |
|---|---|
| High Capital Cost | Reverse Bidding |
| Dust/Soiling | Robotic Dry-cleaning |
| Safiety Materials | Energy Management |
| Low Quality | Energy Factorating |
| Colleg / Reptiration | Energy Alliances |

## SOLUTIONS IMPLEMENTED



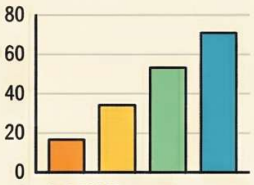
Robotic Cleaning Bots panels

Thermal Management Vents

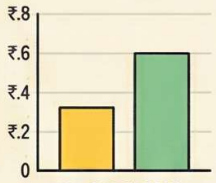
Dedicated Substation

Policy Documents

## MEASURED IMPACT




**~6 Million Tonnes**
$CO_2$ Reduced per Year




**₹2.36/kWh**
Solar Tariff (vs Coal)

**~6,000**
Jobs Created

**~26-28%**
PLF

**Carbon Offset**

## STRATEGIC IMPORTANCE

- Goals in renewable project
- Gobrats and produvement dıions
- Tarerts to inemient economic

## KEY BENEFITS

- Improves station and duration
- Capacitate healith and sciences
- Provides ns of chaviity
- Enuronrer status and environment

## CONCLUSION

Bhadla Solar Park is a summ of global benchmark status, and pernewable renutrant capact is t have therend of wisofar energy economies.